

✅ 📅 DAY 5 — PDF INGESTION + CLEANING PIPELINE

🎯 GOAL

You will build:

PDF → Extract text → Clean → Chunk → Store → Search ✅

👉 So user can upload PDF instead of text file

🧠 WHY THIS IS IMPORTANT

Right now: sample.txt ❌ (demo only)

After today: real PDFs ✅ (industry use)

✅ ✅ STEP 1 — INSTALL PDF LIBRARY

Inside venv:

pip install pdfplumber

👉 Lightweight + works offline ✅

✅ ✅ STEP 2 — CREATE PDF INGESTOR

📁 ingestion/pdf_ingestor.py

✅ FULL CODE (WITH EXPLANATION)

```
import pdfplumber

from ingestion.cleaning import clean_text
from ingestion.chunking import chunk_text

def load_pdf(filepath):

    """ Load PDF and convert to cleaned chunks """

    full_text = ""
```

```
# Open PDF
```

```
with pdfplumber.open(filepath) as pdf:
```

```
    # Loop through pages
```

```
    for i, page in enumerate(pdf.pages):
```

```
        # Extract text from page
```

```
        text = page.extract_text()
```

```
        if text:
```

```
            # Add page marker (optional but useful)
```

```
            full_text += f"\n[Page {i+1}]\n"
```

```
            full_text += text
```

```
# ✅ Step 1: clean text
```

```
cleaned_text = clean_text(full_text)
```

```
# ✅ Step 2: chunk text
```

```
chunks = chunk_text(cleaned_text)
```

```
return chunks
```

WHAT THIS DOES

- ◆ pdfplumber

→ extracts text from PDF

- ◆ cleaning

→ removes:

- extra spaces
- line breaks
- junk

◆ chunking

→ splits into:

500–600 token chunks

👉 needed for RAG ✅

✅ ✅ STEP 3 — UPDATE MAIN TO USE PDF

📁 app/main.py

✅ REPLACE THIS:

```
from ingestion.loader import load_text_file
```

```
filepath = "data/raw/sample.txt"
```

```
docs = load_text_file(filepath)
```

✅ WITH:

```
from ingestion.pdf_ingestor import load_pdf
```

```
filepath = "data/raw/sample.pdf"
```

```
docs = load_pdf(filepath)
```

✅ ✅ STEP 4 — ADD PDF FILE

Place a real PDF in:

data/raw/sample.pdf

👉 Examples:

- technical document
 - manual
 - report
-

✅✅ STEP 5 — RUN SYSTEM

```
python -m app.main
```

✅✅ TEST

Try:

>> summarize the document

>> what is discussed in page 1

>> explain main topic

✅ EXPECTED OUTPUT

👉 System will:

- read PDF ✅
 - clean it ✅
 - chunk it ✅
 - store in DB ✅
 - answer using it ✅
-

🧠 IMPORTANT IMPROVEMENT

Earlier:

txt → RAG

Now:

PDF → full pipeline → RAG ✅🔥

⚠️ KEY LIMITATIONS (NOW)

pdfplumber can:

- ✅ extract text
- ❌ not extract tables well
- ❌ weak for scanned PDFs

👉 We will fix this in upcoming days

✅ ✅ STEP 6 — OPTIONAL DEBUG

Add this:

```
print("Number of chunks:", len(docs))
```

👉 ensures ingestion working

✅ DAY 5 FINAL ARCHITECTURE

PDF

↓

pdfplumber

↓

clean_text()

↓

chunk_text()

↓

BGE embeddings

↓

Vector DB

↓

BM25 + Fusion

↓

BGE reranker

↓

Qwen answer ✅

WHAT YOU BUILD TODAY

- ✓ First real ingestion pipeline
 - ✓ PDF understanding
 - ✓ Document-based QA
-

SYSTEM LEVEL NOW

You moved from:

demo system ✗

to:

- ✓ Document AI system ✓
-

NEXT (DAY 6)

We will improve:

- 👉 OCR (for scanned PDFs & images)
 - 👉 better PDF understanding
 - 👉 table extraction
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